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Rajshahi University of Engineering & Technology

Department of

Electrical & Computer Engineering

Lab Report 3

Implementation of Dynamic Routing using Routing Information Protocol (RIP)

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Implementation of Dynamic Routing using Routing Information Protocol (RIP)

1 Theory and Introduction

Routing is the fundamental process of directing data packets across an interconnected network from a source node to a destination node through various intermediate devices. This experiment focuses on the Network Layer (Layer 3) of the Open Systems Interconnection (OSI) model [1]. While static routing is effective for small, unchanging topologies, it becomes administratively burdensome as networks scale.

Dynamic routing addresses this by allowing routers to automatically exchange information about the network topology. The Routing Information Protocol (RIP) is a distance-vector protocol that utilizes the Bellman-Ford algorithm to determine the best path based on hop count.

In this experiment, RIP was implemented. RIP supports Variable Length Subnet Masking (VLSM) and utilizes multicast addressing for routing updates, which prevents unnecessary processing by non-routing devices on the network [2].

2 Methodology

The experiment was conducted using Cisco Packet Tracer (Version 9.0) [3]. The topology from the previous experiment was extended and converted to a dynamic routing environment through the following steps:

- **Topology Extension:** A fourth router (Router 3) and a corresponding LAN were integrated into the existing network. New serial links were established to connect Router 2 to Router 3.
- **Removal of Static Routes:** All previously configured manual routes were deleted using the `no ip route` command to ensure that the dynamic protocol had full control over the routing table.
- **RIP Activation:** The dynamic routing process was initiated on all routers using the `router rip` command. The version was explicitly set to `version 2`.
- **Network Advertisement:** Each router was configured to "advertise" its directly connected networks using the `network` command. For example, Router 0 advertised 10.0.0.0 and 40.0.0.0.
- **Verification:** Connectivity was verified across all four subnets using the *Ping* and *Tracert* utilities to ensure the routers had successfully synchronized their tables.

Table 1: Extended Network Addressing Table (RIPv2)

Device	Interface	IPv4 Address	Subnet Mask	Default Gateway
Router 0	Fa0/0	10.10.10.0	255.0.0.0	N/A
Router 0	Se2/0	40.40.40.1	255.0.0.0	N/A
Router 1	Fa0/0	20.20.20.0	255.0.0.0	N/A
Router 1	Se2/0	40.40.40.2	255.0.0.0	N/A
Router 1	Se3/0	50.50.50.1	255.0.0.0	N/A
Router 2	Fa0/0	30.30.30.0	255.0.0.0	N/A
Router 2	Se2/0	50.50.50.2	255.0.0.0	N/A
Router 2	Se3/0	70.70.70.1	255.0.0.0	N/A
Router 3	Fa0/0	60.60.60.0	255.0.0.0	N/A
Router 3	Se2/0	70.70.70.2	255.0.0.0	N/A
PC 11	NIC	10.10.10.1	255.0.0.0	10.10.10.0
Laptop 12	NIC	10.10.10.2	255.0.0.0	10.10.10.0
PC 21	NIC	20.20.20.1	255.0.0.0	20.20.20.0
Laptop 22	NIC	20.20.20.2	255.0.0.0	20.20.20.0
PC 31	NIC	30.30.30.1	255.0.0.0	30.30.30.0
PC 32	NIC	30.30.30.2	255.0.0.0	30.30.30.0
PC 41	NIC	60.60.60.1	255.0.0.0	60.60.60.0
Laptop 42	NIC	60.60.60.2	255.0.0.0	60.60.60.0

3 Results

The comprehensive addressing scheme for the extended network is summarized in Table 1.

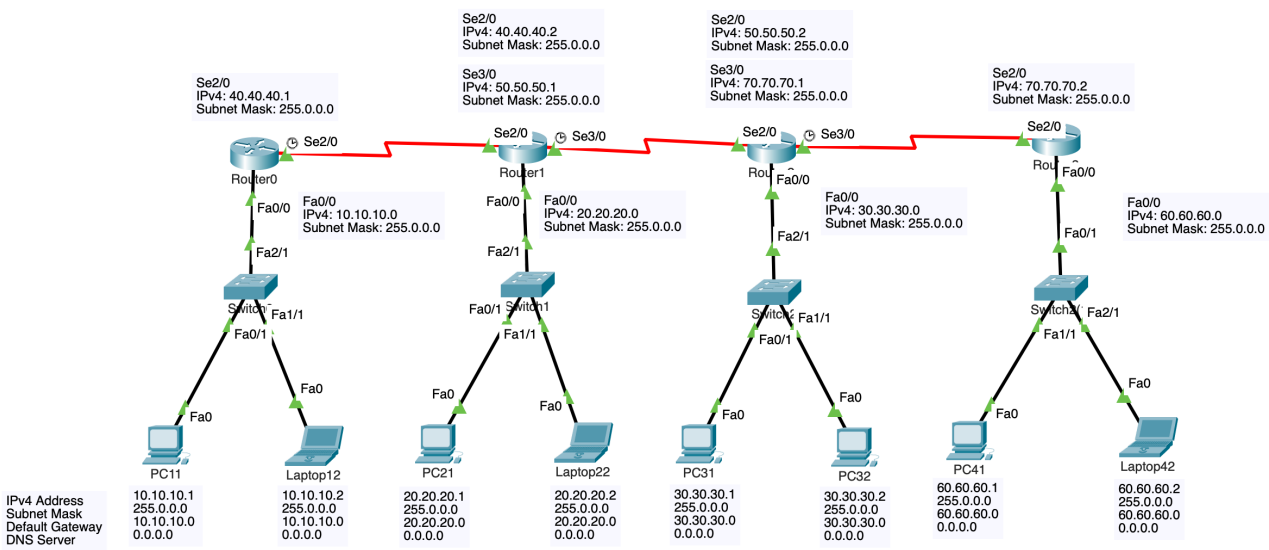

Figure 1: Extended Topology with 4 Routers and Dynamic RIP Routing

```
%SYS-5-CONFIG_I: Configured from console by console

Router#enable
Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

R    10.0.0.0/8 [120/2] via 50.50.50.1, 00:00:07, Serial2/0
R    20.0.0.0/8 [120/1] via 50.50.50.1, 00:00:07, Serial2/0
C    30.0.0.0/8 is directly connected, FastEthernet0/0
R    40.0.0.0/8 [120/1] via 50.50.50.1, 00:00:07, Serial2/0
C    50.0.0.0/8 is directly connected, Serial2/0
R    60.0.0.0/8 [120/1] via 70.70.70.2, 00:00:26, Serial3/0
C    70.0.0.0/8 is directly connected, Serial3/0
```

Figure 2: Routing Table showing RIP Learned Routes (Marked with 'R')

4 Discussion

The primary observation in this experiment was the efficiency of network convergence. Upon adding the 70.0.0.0 network, the information propagated automatically through the neighbors. As demonstrated in Figure 2, the routing table entries were updated to reflect the new topology without manual intervention.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC11	Laptop12	IC...		0.000	N	0	(...	(delete)
	Failed	PC11	Laptop22	IC...		0.000	N	1	(...	(delete)
	Successful	PC11	Laptop22	IC...		0.000	N	2	(...	(delete)
	Failed	Laptop42	PC31	IC...		0.000	N	3	(...	(delete)
	Successful	Laptop42	PC31	IC...		0.000	N	4	(...	(delete)

Figure 3: Verification of End-to-End Connectivity via Successive ICMP Requests

In the *tracert* analysis from PC11 to PC41, the packet traversed three intermediate routers. This confirmed that RIP correctly calculated the shortest path based on the minimum hop count. The "Initial Failure" phenomenon was again observed due to ARP resolution (Figure 3), but subsequent communication was seamless. This experiment highlights that while RIPv2 is limited by a maximum hop count of 15, its ease of configuration makes it superior to static routing for expanding networks.

5 Conclusion

The implementation of dynamic routing using RIPv2 was successfully achieved. The experiment demonstrated the protocol's ability to automate routing table updates and manage multi-hop communication without manual path definition. By comparing this to the previous static routing lab, the scalability and reduced administrative overhead of dynamic protocols were clearly observed.

References

[1] A. S. Tanenbaum, N. Feamster, and D. J. Wetherall, *Computer Networks*, 6th ed. Pearson, 2021.

[2] W. Stallings, *Data and Computer Communications*, 10th ed. Pearson, 2013.

[3] *Cisco Packet Tracer 9.0 User Guide*, Cisco Networking Academy, 2026.